

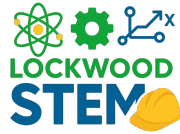
Introduction to Engineering Design

Aerospace Academy Version

Unit 6 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 6

Assemblies, Mechanisms & Aerospace Systems

Multi-part design, assemblies, joints, motion, exploded views, BOMs, assembly instructions, and system documentation.

Grading Rubrics and Scoring Guide

Unit 6: Assemblies, Mechanisms & Aerospace Systems

Assessment Purpose

Multi-part design, assemblies, joints, motion, exploded views, BOMs, assembly instructions, and system documentation. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 6 Score Summary

Category	Points Possible	Points Earned
System purpose and requirements	10	
Part design and fit	15	
Assembly modeling and joints	15	
Mechanism function	15	
Exploded view and assembly documentation	10	
BOM and part naming	10	
Assembly instructions	10	
Testing and iteration	10	
Professional package	5	



Lockwood STEM

Academy of Advanced Aerospace Engineering

Page 3

Unit 6 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete	
System Purpose & Requirements	10	Assembly has a clear aerospace purpose and requirements are specific and testable.	Purpose and requirements are clear.	Purpose or requirements are vague.	No clear purpose.	
Part Design & Fit	15	Parts fit together intentionally with proper clearances, fasteners, interfaces, and manufacturability.	Parts mostly fit with minor issues.	Fit is inconsistent or requires excessive adjustment.	Parts do not fit or are missing.	
Assembly Modeling & Joints	15	CAD assembly uses appropriate components, joints/constraints, and demonstrates intended motion.	Assembly model mostly works with minor joint issues.	Assembly is incomplete or motion is unclear.	No functional assembly model.	Introduction to Engineering Design Aerospace Academy Version
Mechanism Function	15	Mechanism performs intended motion reliably and shows consideration of limits, interference, and safety.	Mechanism works with minor problems.	Mechanism is partially functional or unreliable.	Mechanism does not work.	
Exploded View & Assembly Documentation	10	Exploded view clearly communicates assembly order and part relationships.	Exploded view is mostly clear.	Exploded view is incomplete or confusing.	Missing.	
BOM & Part Naming	10	BOM is complete with clear part names, quantities, materials/processes, and revisions.	BOM is mostly complete.	BOM has missing quantities or unclear names.	No BOM.	
Assembly Instructions	10	Instructions are sequential, clear, include checks, and could be followed by another team.	Instructions are usable with minor gaps.	Instructions are unclear or missing steps.	No instructions.	
Testing & Iteration	10	Assembly is tested, issues are documented, and improvements are evidence-based.	Testing and revision notes are adequate.	Testing is superficial or revisions unclear.	No test evidence.	Rubrics
Professional Package	5	Final system package is organized and presentation ready.	Package is complete.	Package is incomplete.	Missing.	

Student Self-Check Before Submission

	Requirement	Complete?
☐	System purpose and requirements evidence is complete and ready to be graded.	
☐	Part design and fit evidence is complete and ready to be graded.	
☐	Assembly modeling and joints evidence is complete and ready to be graded.	
☐	Mechanism function evidence is complete and ready to be graded.	
☐	Exploded view and assembly documentation evidence is complete and ready to be graded.	
☐	BOM and part naming evidence is complete and ready to be graded.	
☐	Assembly instructions evidence is complete and ready to be graded.	
☐	Testing and iteration evidence is complete and ready to be graded.	
☐	Professional package evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
those b